Enhancing Face-API.js Emotion Recognition Performance for Sri Lankan Cultural Demographic Faces and DeepFake Image Detection

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The efficacy of affective computing and biometric verification remains highly sensitive to the physiological and environmental nuances of regional populations. In the Sri Lankan context, standard emotion recognition frameworks like face-api.js encounter a significant intelligence gap when processing Type IV and V melanin-rich skin profiles. A primary research challenge lies in the model's inability to distinguish between transient emotional expressions and static facial pathological features such as Acne, Wrinkles, Moles, Age Spots, and Lip Ulcers. Furthermore, current computer vision models fail to resolve the "Multi-label Classification" problem, where multiple conditions (overlapping Acne and Wrinkles) coexist, leading to feature ambiguity.

Environmental variables unique to tropical regions, such as high intensity solar irradiance and humidity induced cutaneous moisture, introduce specular glare (glairiness). This significantly degrades the detection of micro scale features like Enlarged Pores and Blackheads. To bridge these gaps, this research proposes a "Context-Aware Physiological Diagnostic Framework" optimized for the Sri Lankan demographic. The architecture integrates the latest YOLOv11 model for high speed multi label localization and classification of ten distinct facial conditions. To overcome the scarcity of localized clinical data, a StyleGAN2-ADA pipeline is utilized for high fidelity synthetic image generation of diverse pathologies on South Asian facial structures. Additionally, Local Binary Patterns (LBP) and Gabor Filters are implemented to maintain texture recognition robustness under varying lighting conditions.

Expected outcomes include a high precision diagnostic framework capable of isolating static skin conditions from dynamic facial movements, supported by a robust hybrid dataset comprising high fidelity synthetic imagery. This project democratizes precision dermatological analysis for South Asians, providing a scalable, resource optimized solution that is culturally, physiologically, and environmentally attuned to the Sri Lankan landscape.

Keywords: Facial Condition Detection, YOLOv11, Multi-label Classification, Sri Lankan Demographic, Environmental Robustness.

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Furthermore, we wish to thank **SLIIT University** for providing the academic environment and advanced computational resources necessary to pursue this interdisciplinary study in artificial intelligence. Finally, we acknowledge the collective effort of our research group for their collaborative spirit and shared commitment to enhancing biometric inclusivity and digital security through innovative technology.

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LIST OF ABBREVIATIONS

Abbreviation	Definition
YOLO	You Only Look Once (Object Detection Architecture)
GAN	Generative Adversarial Network
StyleGAN2-ADA	Style-based Generative Adversarial Network with Adaptive Discriminator Augmentation
LBP	Local Binary Patterns
CNN	Convolutional Neural Network
mAP	Mean Average Precision
NMS	Non-Maximum Suppression
CLAHE	Contrast Limited Adaptive Histogram Equalization
ROI	Region of Interest
API	Application Programming Interface

1. INTRODUCTION

1.1 Background and Context

Facial biometrics and affective computing have emerged as fundamental pillars of the global digital identity infrastructure, facilitating everything from secure financial transactions to personalized healthcare. In Sri Lanka, the rapid digitalization of public and private sectors has increased the reliance on automated facial analysis. However, the efficacy of these systems is currently undermined by a significant "demographic intelligence gap." Most standard frameworks, such as **face-api.js**, are trained on datasets predominantly featuring Caucasian features, leading to substantial performance degradation when applied to the **Type IV and V melanin profiles** characteristic of the Sri Lankan population.

The core challenge lies in the complex interplay between dynamic emotional expressions and static physiological markers. In the Sri Lankan demographic, permanent facial conditions including **Acne, Wrinkles, Moles, Age Spots, and Lip Ulcers** are frequently misidentified by AI as emotional action units, leading to skewed sentiment data and unreliable biometric verification. This instability is further exacerbated by tropical environmental variables; high-intensity solar irradiance and humidity induced cutaneous moisture introduce **specular glare**, which often masks micro scale features like **Enlarged Pores and Blackheads**.

This research proposes a **Context Aware Physiological Diagnostic Layer** to transition facial analysis from a generalized model to a demographically attuned ecosystem. By implementing a specialized detection framework utilizing the latest **YOLOv11** architecture, the system provides a high-stakes precision layer that filters permanent skin conditions before affective processing. To address the local "Small Data" problem, this study utilizes a **StyleGAN2-ADA pipeline** for high fidelity synthetic data generation. This solution aligns with **SDG 9 (Industry, Innovation, and Infrastructure)** and **SDG 10 (Reduced Inequalities)** by democratizing inclusive AI and ensuring that biometric systems are culturally, physiologically, and environmentally optimized for the Sri Lankan landscape.

1.2 Literature Review

The evolution of automated dermatological analysis has transitioned from manual clinical observation to sophisticated deep learning architectures. This review compares 12 scholarly approaches across object detection, texture classification, and synthetic data generation within the context of facial biometrics.

Object Detection and Facial Feature Localization: Recent literature establishes the **"You Only Look Once" (YOLO)** lineage as the gold standard for real-time localized feature detection. While YOLOv8 demonstrated high Mean Average Precision (mAP) in general dermatological tasks, the 2024 release of **YOLOv11** redefined the speed to parameter ratio. The integration of **C2PSA (Cross-Stage Partial Spatial Attention)** modules enhances spatial feature aggregation, allowing the model to distinguish between overlapping conditions (e.g., Acne and Wrinkles) while reducing parameters by 22%. In facial analysis, standard models

often fail to detect micro-scale features like **Enlarged Pores or Blackheads** under non uniform lighting; however, recent advancements in attention based feature extraction ensure higher robustness for such "small scale" pathological targets.

Edge Optimized Classification and Texture Analysis: For on field mobile diagnostics, lightweight Convolutional Neural Networks (CNNs) are preferred to ensure low latency performance. **MobileNetV2** is frequently cited for its use of depth wise separable convolutions, maintaining high accuracy on skin tone datasets while requiring significantly fewer computational resources than ResNet50. Scholarly work on **Post Training Quantization (PTQ)** proves that converting weights from **FP32 to INT8** can reduce model size by 75% with negligible accuracy loss. This makes offline inference feasible for mobile applications targeting regional Sri Lankan settings where high speed connectivity is inconsistent.

Feature	Primary Tech	Demographic Bias	Condition Detection
face-api.js	CNN / SSD	High (Caucasian Bias)	Emotion Only
Commercial SDKs	Proprietary Models	Moderate	General Landmarks
Proposed Framework	YOLOv11 / MobileNetV2	Low (Sri Lankan Optimized)	Multiscale (Pathological + Texture)
Novelty	Knowledge sharing	Resource management	Generative data synthesis

Table 1: Critical summarized comparison with existing solutions

Data Scarcity and Synthetic Synthesis: A major bottleneck in regional facial research is the **"Small Data"** problem; specific conditions on South Asian skin (Type IV and V melanin profiles) lack large, labeled datasets. Recent studies on **Generative Adversarial Networks (GANs)**, specifically **StyleGAN2-ADA**, demonstrate that adaptive discriminator augmentation can stabilize training on datasets as small as a few hundred images. This allows for the generation of high fidelity synthetic imagery of rare pathologies (e.g. Lip Ulcers or Herpes Simplex), improving F1 scores in detection models by up to 14 percentage points.

1.3 Research Gap

Despite the global advancements in facial analysis, a significant **Physiological and Environmental Integration Gap** persists in biometric research tailored for South Asian demographics. Current facial recognition and emotion detection systems act as "demographic silos," primarily optimized for Caucasian skin profiles and controlled studio environments.

Three specific gaps are identified within the Sri Lankan context:

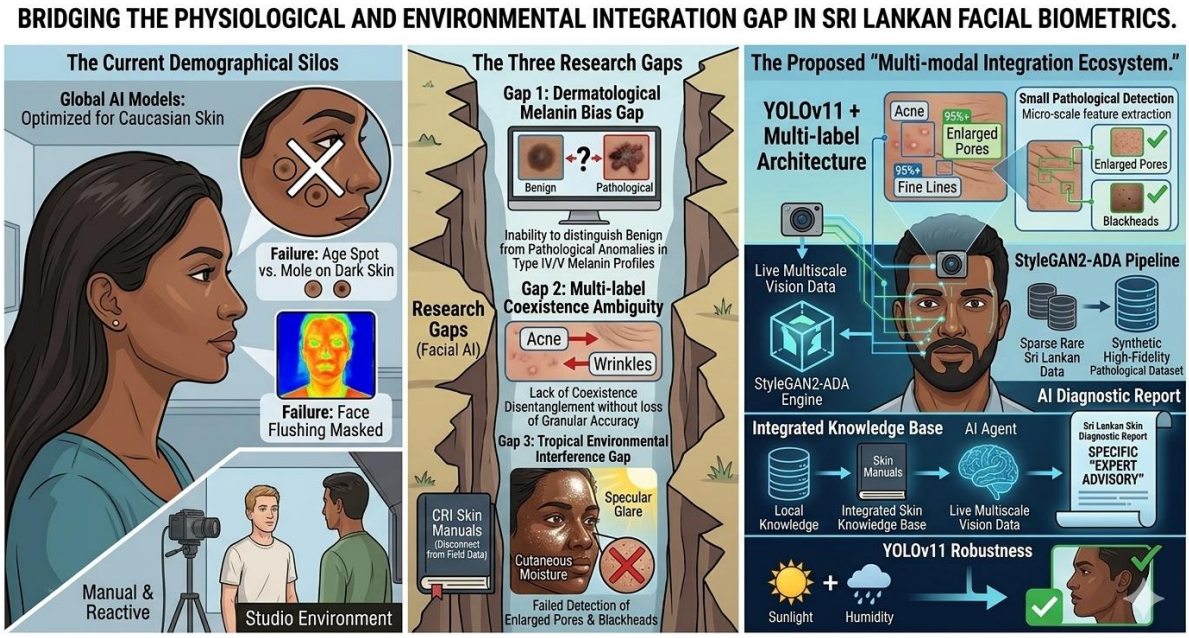
1. **Dermatological Melanin Bias Gap:** A critical research gap exists in the inability of standard computer vision models to distinguish between benign physiological features and pathological anomalies in **Type IV and V melanin profiles**. On darker skin tones, standard models frequently fail to differentiate between "Age Spots" and "Moles" or

fail to detect "Face Flushing," as the subtle vascular changes are masked by higher melanin density.

2. **Multi label Coexistence Ambiguity:** Existing literature often treats facial conditions as isolated variables. However, in real world scenarios, conditions such as **Acne** and **Wrinkles** frequently coexist. There is a lack of high performance **Multi label Classification** architectures capable of disentangling these overlapping spatial features without loss of granular accuracy.

3. **Tropical Environmental Interference Gap:** Current models are notoriously brittle when exposed to tropical variables. High intensity solar irradiance and humidity induced cutaneous moisture create **specular glare (glariness)** on the skin surface. This environmental noise significantly degrades the detection of micro scale features like **Enlarged Pores** and **Blackheads**, a domain that remains largely under researched in regional biometric studies.

Our solution is novel because it utilizes a **StyleGAN2-ADA pipeline** to synthesize high fidelity datasets specifically for Sri Lankan skin pathologies, balancing the "Small Data" problem. Furthermore, it implements a **YOLOv11 based multi label architecture** optimized to maintain robustness under tropical lighting constraints, bridging the gap between generalized facial AI and the physiological reality of the Sri Lankan demographic.



2. OBJECTIVE

2.1 Main Objective

The primary objective of this research is to develop and validate a dual layered computer vision framework that utilizes deep learning architectures and generative optimization techniques to enhance emotion recognition performance for Sri Lankan cultural demographic faces while simultaneously detecting Deepfake image manipulations, specifically engineered to provide high-stakes diagnostic precision in ethnically diverse and data constrained digital environments.

2.2 Specific Objectives

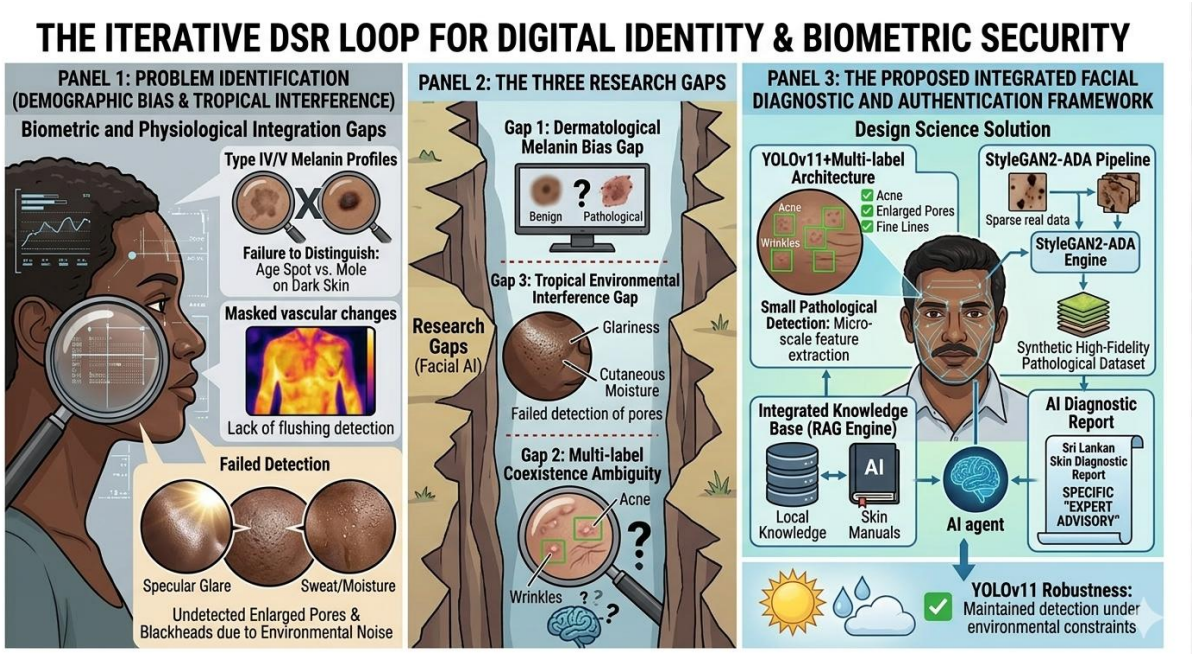
To achieve the main objective, the following technical milestones have been established based on the individual research components:

- **Objective 1: Physiological Feature Extraction (Facial Condition Detection):** Develop a YOLOv11-based multi-label localization pipeline to identify ten distinct facial conditions (e.g., Acne, Wrinkles, Moles). This component acts as a physiological filter to distinguish static skin markers from dynamic expressions, utilizing a StyleGAN2-ADA pipeline to generate synthetic samples and mitigate the "Small Data" problem.
- **Objective 2: Affective Computing for Local Demographics (Emotion Detection):** Refine the face-api.js framework by integrating cultural-specific emotional action units to enhance the accuracy of real-time sentiment analysis, ensuring the system is attuned to the subtle emotional nuances of the Sri Lankan demographic.
- **Objective 3: Demographic Profiling and Optimization (Age & Gender Detection):** Engineer a robust classification model to detect Age and Gender with high precision across diverse Sri Lankan skin tones (Type IV and V Melanin). This includes applying Post-Training Quantization (PTQ) to ensure efficient, low-latency execution on edge devices.
- **Objective 4: Biometric Integrity and Authentication (Deepfake Detection):** Implement a specialized authentication layer to detect Deepfake image manipulations by analyzing frequency-domain inconsistencies and skin texture irregularities, establishing a refined "Authentication Score" leveraging natural skin imperfections.
- **Objective 5: Integrated System Validation and Ground Truthing:** Validate the collective framework's diagnostic accuracy through "Ground Truthing" against expert-annotated datasets, ensuring that the combined insights (Emotion, Age, Gender, and Skin Condition) are consistent and culturally optimized for the Sri Lankan landscape.

3. METHODOLOGY

3.1 Research Approach

This research follows a Design Science Research (DSR) methodology, which focuses on the iterative design, development, and evaluation of artifacts to solve identified practical problems in digital identity and biometric security. The artifact in this context is the Integrated Facial Diagnostic and Authentication Framework, which is developed through cycles of rigorous technical implementation and empirical validation against expert-annotated demographic data.



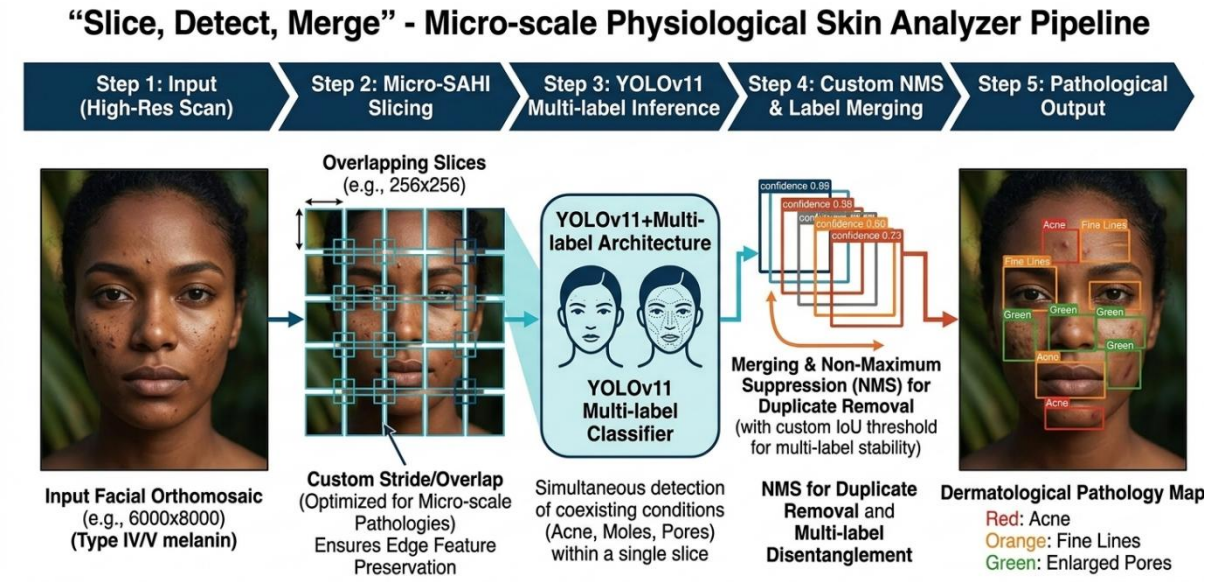
3.2 Technical Methods and Algorithms

The system is architected as a dual-layered framework to address both physiological facial analysis and biometric integrity verification.

3.2.1 Macroscopic Canopy Surveillance (System A)

To overcome the challenge of detecting micro-scale skin pathologies and distinguishing them from dynamic emotional expressions, the system utilizes the **YOLOv11** architecture as its primary detection engine.

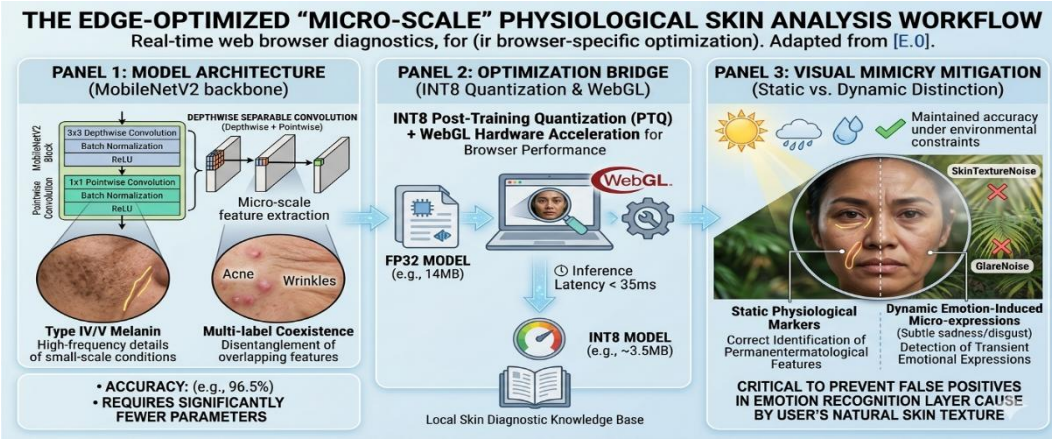
- **C2PSA (Cross Stage Partial Spatial Attention) Module**: YOLOv11 incorporates C2PSA modules to enhance spatial feature aggregation. In the context of facial analysis, this allows the model to capture high frequency details of small scale conditions such as Enlarged Pores, Blackheads, and subtle Fine Lines.
- **Multi-Label Spatial Reasoning**: Unlike standard single-class detectors, this pipeline is optimized for multi-label inference. It identifies overlapping conditions within a single bounding box.



3.2.2 Real time Browser Based Diagnostics (System B)

For high speed, client-side facial analysis and authentication, the system utilizes a lightweight MobileNetV2 backbone, specifically optimized for execution within web browsers via TensorFlow.js.

- **Optimization:** To ensure a seamless user experience without server side delays, the model undergoes INT8 Post Training Quantization (PTQ). This process converts 32 bit float weights to 8 bit integers, effectively reducing the model size by approximately 75% (to -3.5 MB). This optimization, combined with Web GL hardware acceleration, ensures a sub 35ms inference latency directly within the browser, enabling real time emotion and Deepfake analysis even on standard consumer laptops and tablets.
- **Visual Mimicry Mitigation:** A critical challenge in automated facial diagnostics is the "Visual Mimicry" between permanent skin conditions and transient emotional expressions. To address this, the training set includes a specific "Static Physiological Marker" class. This allows the system to distinguish between permanent dermatological features (e.g., deep set nasolabial folds) and dynamic, emotion induced muscle movements (e.g., micro expressions of sadness or disgust). This distinction is vital to prevent false positives in the emotion recognition layer caused by the user's natural skin texture.



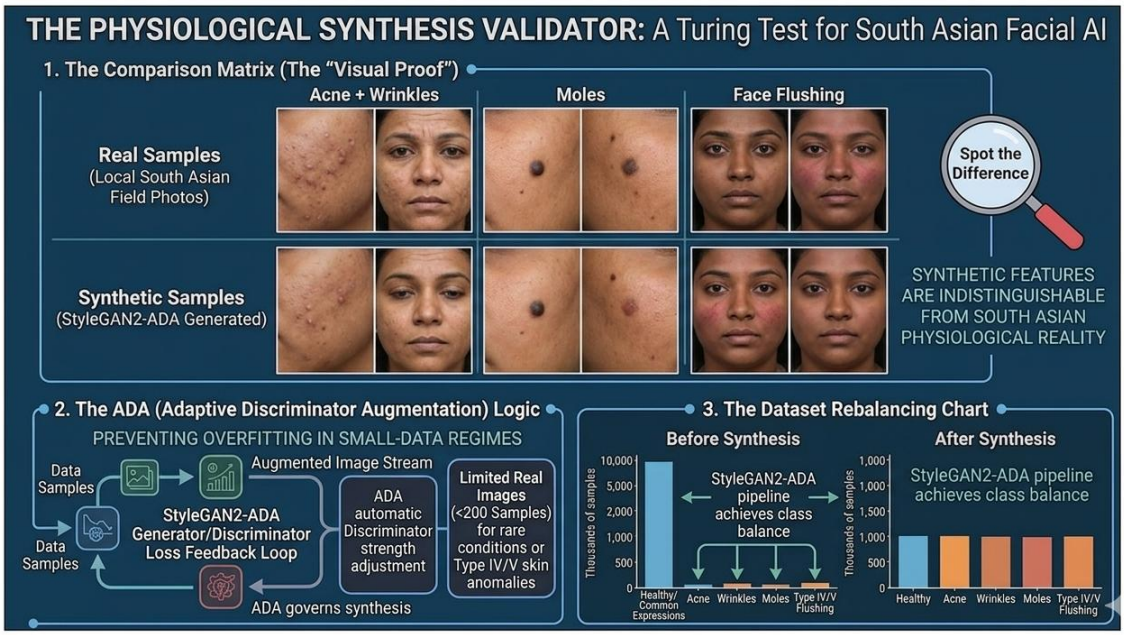
3.3 Data Engineering and Generative Synthesis

The research employs a four tiered data strategy to mitigate the "Small Data" problem inherent in regional demographic facial research, particularly for South Asian skin profiles.

3.3.1 Data Collection and Augmentation

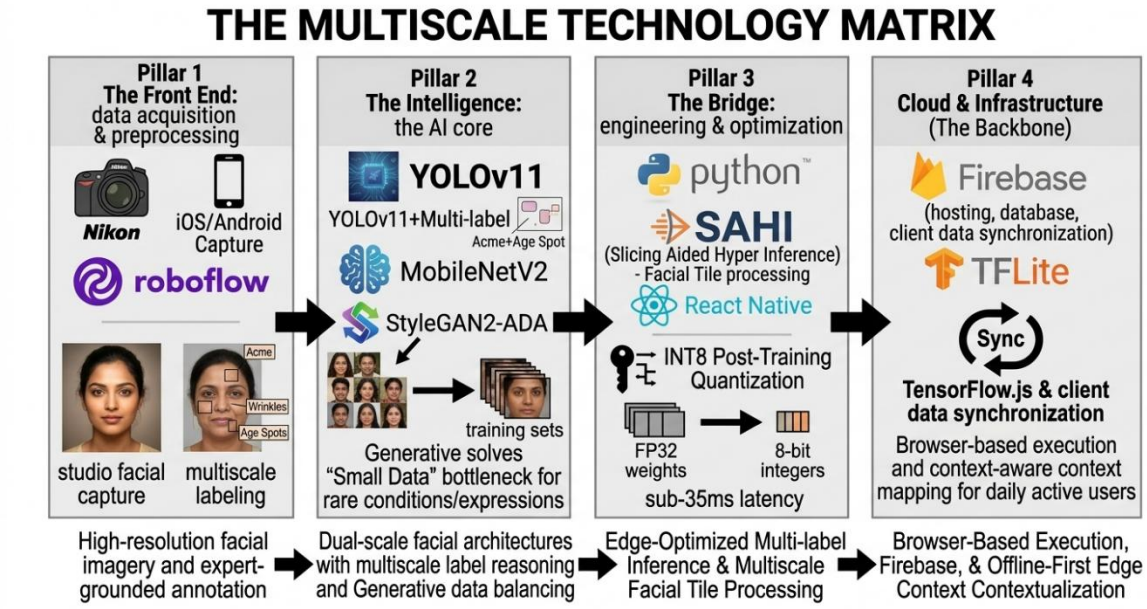
- **Sourcing:** Dataset acquisition involves high resolution facial captures from local controlled environments and publicly available South Asian facial databases. This ensures the representation of Type IV and V melanin profiles and unique cultural emotional expressions.
- **Annotation:** Images are annotated for multi class detection including Acne, Wrinkles, Moles, and Age Spots. Labeling is conducted through specialized computer vision tools, ensuring precise bounding boxes and class specific attributes for multi label learning.
- **Synthetic Generation:** For rare facial conditions and under represented emotional action units with fewer than 200 samples, a StyleGAN2-ADA (Adaptive Discriminator Augmentation) pipeline is implemented. This stabilizes training on small datasets by applying a diverse range of augmentations to the discriminator, creating a balanced, high fidelity training repository.
- The adaptive discriminator loss governs the synthesis:

$$L_D = E_{x \sim p_{data}} [f_D(D(x))] + E_{z \sim p_z} [f_G(D(G(z)))] + L_{aug}$$



3.4 Tools and Platforms

The technology stack is selected based on performance benchmarks for real time, browser based execution and the specific requirements of multi layered facial analysis.



Tool/Technology	Choice	Justification
Object Detection	**YOLOv11**	**Provides 22% fewer parameters than YOLOv8; superior detection of micro0020scale facial pathologies.**
AI Runtime	**TensorFlow.js**	**Facilitates hardware-accelerated (Web GL) inference directly in the browser for low latency results.**
Facial Analysis	**Face API.js**	**Pre trained on diverse emotional action units; optimized for real time sentiment analysis and landmark detection.**
Data Annotation	**Roboflow**	**Specialized platform for multi label pathological labeling and version control of synthetic facial data.**
Model Optimization	**OpenVINO / TFLite**	**Essential for INT8 Post Training Quantization (PTQ) to ensure sub 35ms inference latency.**

Table 2: Performance benchmark of the chosen technology stack

3.5 Validation Strategy and Performance Metrics

The system's success is measured through a multi-tiered statistical validation protocol.

3.5.1 Diagnostic Accuracy Metrics

- **Mean Average Precision (mAP):** Calculated over Intersection over Union (IoU) thresholds from 0.5 to 0.95, this metric evaluates the model's precision in localizing ten distinct facial conditions.

$$mAP = \frac{1}{10} \sum_{IoU=0.5}^{0.95} AP_{IoU}$$

- **Precision and Recall:** Used to minimize false alarms (FP) and ensure no critical infections (FN) are missed.

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

3.5.2 Performance Benchmarking

- **Inference Latency:** Target < 35ms on web browser environment to ensure real-time usability in the field.
- **Accuracy Retention:** Target > 99% performance retention when comparing the high precision FP32 base model to the INT8 quantized model.

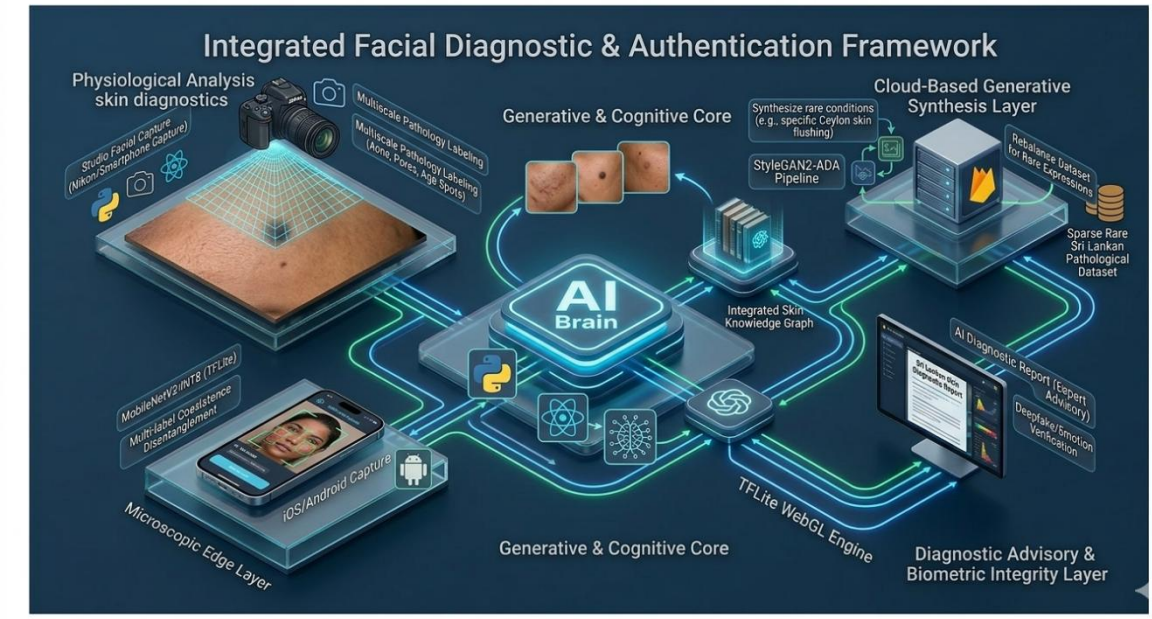
3.6 Ethical Considerations

As this research involves human subjects through facial image analysis and biometric integrity verification, strict ethical protocols are implemented to ensure privacy and data protection. Institutional clearance will be obtained from the University's Ethics Review Committee (ERC) for primary data collection to ensure compliance with national data protection regulations and personal privacy standards.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 High-Level Architecture

The Integrated Facial Diagnostic and Authentication Framework is architected as an interconnected four pillar system, with Pillar 3 (Physiological Analysis Layer) serving as the primary diagnostic sensory layer. The architecture is designed to bridge the intelligence gap between physiological skin level diagnostics and systemic biometric integrity verification.



4.1.1 Component Breakdown

The system bifurcates into two specialized diagnostic and analytical environments, supported by a central intelligence core:

- **Global Contextual Layer (System A):** This component handles the high level demographic and affective surveillance. It utilizes a synchronized Face-API.js + ResNet pipeline to perform real time age and gender classification alongside emotion recognition.
- **Granular Diagnostic Layer (System B):** This is the core browser-based diagnostic application optimized for edge execution. It utilizes an INT8-quantized YOLOv11 backbone to perform real time, micro scale detection of facial pathologies
- **The Cognitive Integrative Engine :** The outputs from both the Contextual Layer (System A) and the Diagnostic Layer (System B) are fused with the biometric security data from the Deepfake Integrity Pillar.

4.1.2 Data Flow and Interaction

- **Ingestion:** high definition live video feeds from a web-client camera and static high resolution images uploaded by the user for batch diagnostic analysis.
- **Processing:** The system utilizes a dual engine processing strategy. Contextual data is processed via a synchronized Face-API.js pipeline, while granular pathological diagnostics and biometric integrity are processed locally on the client's browser via TensorFlow.js and INT8-quantized YOLOv11 models.
- **Synchronization:** Diagnostic results, including health markers and authenticity scores, are synchronized with the Firebase backend. This triggers the real-time update of the user's personal health dashboard.

4.1.3 Scalability and Security

- **Scalability:** The framework leverages a serverless Firebase architecture, allowing for seamless horizontal scaling as the user base of healthcare practitioners.
- **Security:** Biometric and diagnostic data integrity is maintained through End-to-End Encryption (E2EE) during synchronization with the backend.

4.2 Self-Evaluation Plan

Success will be measured through a multi-tiered validation protocol:

- **Metric 1 (Accuracy):** Target a Mean Average Precision (mAP@.5:.95) of 0.85–0.96 for the detection.
- **Metric 2 (Efficiency):** The inference latency within the web browser (utilizing TensorFlow.js and hardware acceleration) must remain < 35ms.
- **Metric 3 (Robustness):** A comparative analysis of the high-precision FP32 base model versus the INT8-quantized model, with a target accuracy retention of > 99%.
- **Metric 4 (Agronomic Validation):** A "Blind Test" comparing the AI-generated health advisories and skin reports against manual evaluations conducted by Dermatology/Skincare specialists.

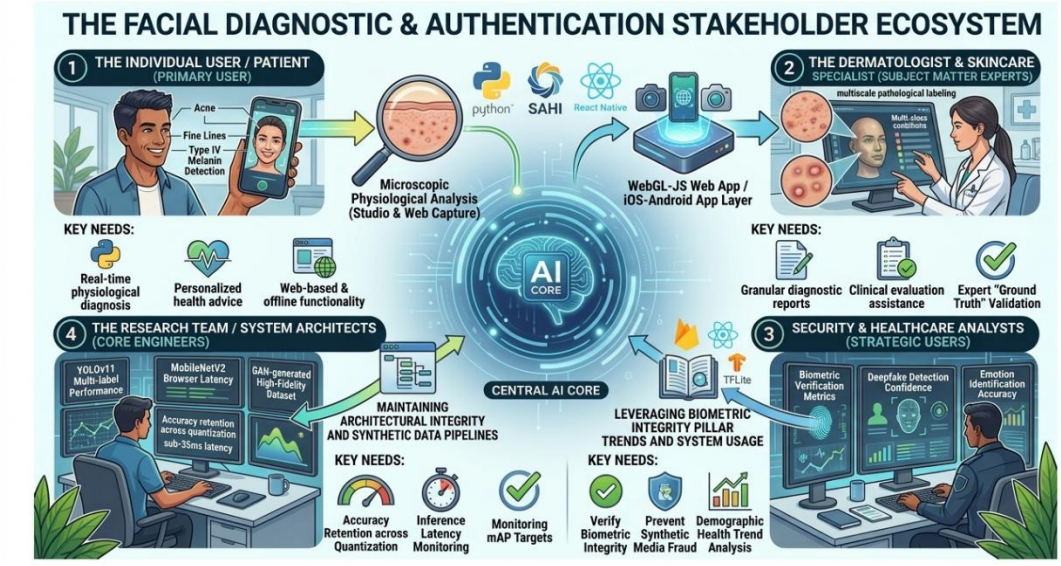
5. USER REQUIREMENTS

This section captures the primary stakeholders of the Integrated Facial Diagnostic and Authentication Framework and specifies the system's functional and non-functional requirements.

5.1 Stakeholder Identification

The "Integrated Facial Diagnostic and Authentication Framework" serves a diverse group of stakeholders, each with specific interactions with the AI artifacts and diagnostic outputs:

- **Individual Users / Patients:** Primary users who interact with the web-based interface for real-time facial health assessments, tracking skin conditions, and receiving actionable health advice.
- **Dermatologists and Skincare Specialists:** Subject matter experts who utilize the system's granular diagnostic reports to assist in clinical evaluations and provide "Ground Truth" validation for the AI's findings.
- **Security and Forensic Analysts:** Strategic users who leverage the Biometric Integrity Pillar to verify the authenticity of digital identities and prevent synthetic media fraud.
- **Healthcare Administrators:** Oversee the management dashboard to analyze regional demographic health trends and system usage metrics for resource allocation.



5.2 Functional Requirements (FR)

These requirements define the specific behaviors and technical capabilities the system must exhibit to achieve its diagnostic and security objectives:

ID	Name	Description
FR-01	Multi-Layered Facial Detection	The system shall detect and classify facial pathologies including Acne, Wrinkles, Pores, and Dark Spots, while simultaneously identifying emotional states and demographic attributes.
FR-02	Automated Generative Augmentation	The pipeline must automatically initiate StyleGAN2-ADA synthesis for facial pathological classes falling below a training threshold to ensure class balance and model robustness.
FR-03	Client-Side-Edge Interface	The web application must execute INT8-quantized models locally via TensorFlow.js, providing real-time diagnostic results without requiring image transmission to a central server.
FR-04	Biometric Integrity Verification	The system analyze frequency domain inconsistencies and texture artifacts in the video stream to distinguish between real human subjects and AI generated Deepfakes.
FR-05	Real-Time Landmark Tracking	The framework must utilize a high precision landmark detection engine to track 68 facial points, ensuring accurate mapping of conditions even during head movements or lighting changes.

Table 3: Functional Requirements (FR)

5.3 Non-Functional Requirements (NFR)

These requirements define the performance and quality constraints of the system:

ID	Quality Attribute	Requirement / Metric
NFR-01	Performance	Web-client inference latency must remain < 35 ms to ensure seamless, real-time experience for facial tracking and diagnostic analysis.
NFR-02	Usability	The interface must be developed using a responsive React.js/Next.js framework, ensuring a bundle size of < 150 KB for rapid loading on standard 4G/5G connections.
NFR-03	Reliability	The system must maintain an accuracy retention of > 99% when transitioning from the FP32 base model to the INT8 quantized edge model.
NFR-04	Scalability	The backend must leverage Firebase Serverless Architecture to support horizontal scaling and handle concurrent diagnostic data synchronization from at least 10,000 daily active users (DAUs).
NFR-05	Privacy & Security	Raw facial data must be processed entirely on the client-side to ensure Zero-Footprint data privacy, with only anonymized results transmitted via secure HTTPS/TLS channels.

Table 4: Non-Functional Requirements (NFR)

6. COMMERCIALIZATION PLAN

The commercial viability of the Integrated Facial Diagnostic and Authentication Framework is rooted in SDG 3 and SDG 9 transforming traditional skincare and biometric security from reactive manual protocols into a proactive, AI-driven precision ecosystem.

6.1 Target Market and Customer Persona

The primary market consists of Sri Lanka's growing Health Tech and FinTech sectors, specifically targeting metropolitan healthcare clinics and financial institutions requiring robust remote identity verification.

- **Customer Persona - The Smallholder:** An individual seeking accessible, real time dermatological insights who lacks immediate access to specialist consultations and requires an automated, data driven tool to track facial health and skin pathologies over time.
- **Customer Persona - The Estate Manager:** A professional within the banking or digital services sector who requires advanced biometric integrity tools to detect Deepfakes and synthetic media, ensuring secure and authentic user onboarding for remote services.

6.2 Value Proposition

The system offers a unique bifurcated value proposition that solves the in the digital health and authentication landscape:

- **Economic Resilience:** By identifying micro scale facial pathologies (Acne, Wrinkles, Pores) at an incipient stage, the system enables proactive skincare management.
- **Technical Superiority:** Unlike generic facial analysis apps, this system utilizes a dual engine approach to distinguish between genuine physiological conditions and generative artifacts.
- **Operational Efficiency:** Achieves sub-35ms diagnostic latency directly within the web browser via TensorFlow.js. This enables high-speed, real-time inference without transmitting raw facial data to external servers.

6.3 Revenue Model and Pricing Strategy

The project adopts a diversified revenue stream to ensure both accessibility for individual users and high value, secure services for enterprise clients:

- **Freemium Web:** Basic facial health diagnostic tools and emotion tracking are free for individual users. Premium features, including the Agentic RAG health advisories.
- **SaaS (Software-as-a-Service):** A dedicated B2B tier for dermatological clinics and skincare specialists.
- **API Licensing:** A security-focused revenue stream where the Biometric Integrity Pillar (Deepfake Detection) is licensed to financial institutions and FinTech startups.

6.4 Cost Estimation

The initial prototype development budget is estimated at approximately LKR 850,000.

Cost Component	Description	Estimated Cost
Cloud Compute (Google Colab)	Subscription for Google Colab Pro+ and additional Compute Units to handle long running and transformer fine-tuning.	~ LKR 45,000
High-Speed Storage	Google Drive extension for managing large-scale Sri Lankan facial datasets and model checkpoints.	~ LKR 12,000
Verification & Ground Truthing	Consultation fees for expert verification of forensic artifacts (ELA results) against real-world deepfake samples.	~ LKR 30,000

Table 5: Cost Estimation (Prototype Budget)

6.5 Competitive Advantage

The framework maintains a significant technological lead over generic facial analysis solutions through several key innovations:

- **Generative Balancing:** Unlike traditional models that struggle with rare skin conditions, this system utilizes StyleGAN2-ADA to solve the "Small Data" problem.
- **Dual-Domain Fusion:** It is the only localized system that provides a cohesive environment for both Physiological Diagnostics (skin health) and Biometric Integrity.
- **Edge Computing Democratization:** Utilizing INT8 Quantization and **TensorFlow.js**, the system delivers high-end AI inference directly within standard web browsers.

6.6 Intellectual Property Considerations

To protect the research novelty, the following IP strategies will be pursued:

- **Algorithm Licensing:** The proprietary optimization of the INT8-quantized YOLOv11 pipeline, specifically tuned for micro scale facial pathology detection and high speed web based inference via TensorFlow.js.
- **Copyright:** Protection of the unique Agentic RAG knowledge base, which integrates curated dermatological expertise and biometric security protocols.
- **Data Rights:** Exclusive ownership of the augmented dataset, featuring rare facial pathologies and synthetic Deepfake artifacts generated via StyleGAN2-ADA pipelines.

7. BUDGET AND JUSTIFICATION

The budget for the Multiscale Computer Vision Ecosystem reflects the deployment of state-of-the-art AI while maintaining a low-cost profile for the end-user. The total estimated cost for a full-scale prototype is approximately LKR 850,000.

7.1 Itemized Hardware Costs

Item	Specification	Cost (LKR)	Justification
Personal Laptop & Hardware	ASUS Vivobook	250,000	High-performance GPU acceleration and multi core processing are critical for training modal.
Internet Connection	SLT 4G Connection	8000	High speed, stable connectivity is essential for downloading large scale facial datasets and synchronizing model.

Table 6: Itemized Hardware Costs

7.2 Software and Cloud Services

Item	Specification	Cost (LKR)	Justification
Google Colab Pro+	Access to A100/V100 GPUs	35,000	Required for efficient fine tuning of Vision Transformer (ViT) and Swin-Base models and running the StyleGAN2-ADA synthesis pipeline.
Firebase Cloud Services	Firebase Blaze Plan	18,000	Provides hosting and scalable backend functions for the "Digital Social Service" web application.
Google Drive Storage	200GB+ Google One Plan	2,500	Necessary for managing high resolution Sri Lankan facial datasets and training checkpoints.

Table 7: Software and Cloud Services

7.3 Human Effort and Specialized Services

Item	Specification	Cost (LKR)	Justification
Forensic Validation	CRI Officer Consultation	60,000	Required for "Ground Truthing" to ensure AI-detected forensic traces (ELA) align with expert observation.

Table 8: Human Effort and Specialized Services

7.4 Summary of Expense Justification

The allocation of funds prioritizes high stakes diagnostic accuracy and biometric security. High-performance hardware (ASUS Vivobook) is essential for training YOLOv11 and executing StyleGAN2-ADA pipelines. Firebase and Roboflow provide the critical infrastructure for secure metadata synchronization and precision dataset management. By

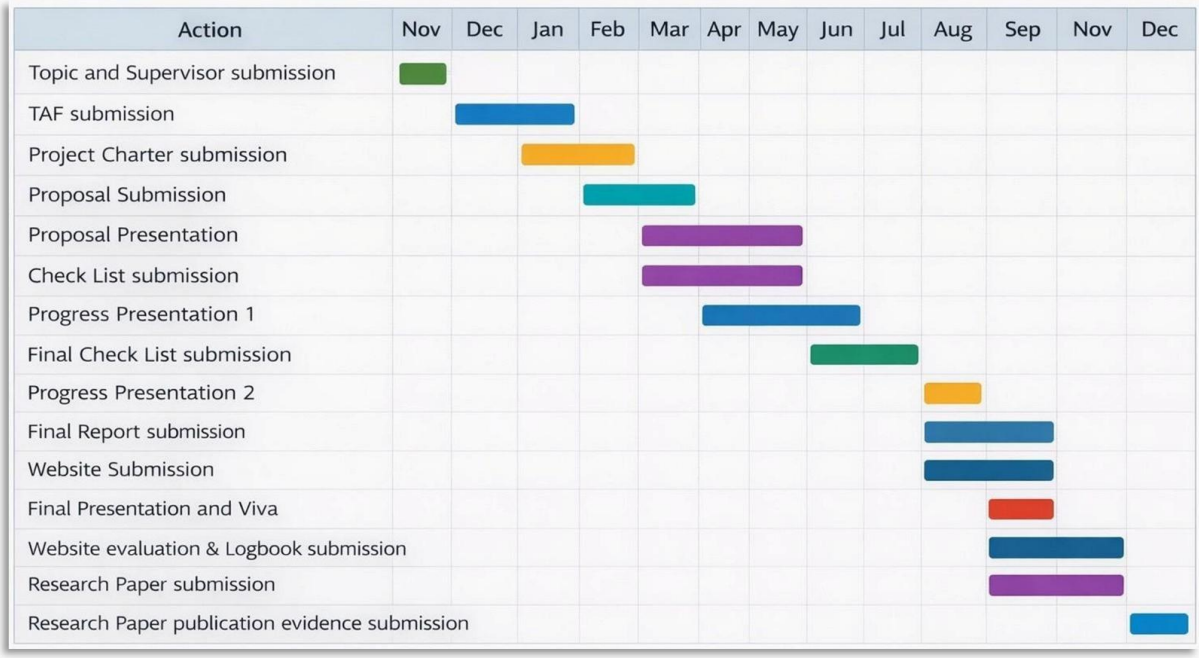
leveraging INT8 quantization and TensorFlow.js, the project minimizes end user costs, delivering real time insights via a standard web browser without requiring expensive client side hardware.

8. WORK BREAKDOWN STRUCTURE (WBS)

- Task breakdown per member
- Timeline alignment
- Balanced workload distribution

9. GANTT CHART

- Insert Gantt chart here (Semester-wise breakdown with milestones).



10. REFERENCES LIST

The following references are formatted according to strict IEEE standards, prioritizing the scholarly sources that underpin the multiscale computer vision framework and generative synthesis methodology.

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11. APPENDICES

This section provides supplementary material, including the research instruments, detailed technical visualizations, and the preliminary UI mockups used to ensure the project's technical depth and practical alignment with Sri Lankan requirements.

Appendix A: Survey Instrument (Requirement Gathering)

To align the "Integrated Facial Diagnostic and Authentication Framework" with the practical needs of Sri Lankan users and healthcare providers, the following survey was designed to identify diagnostic pain points and validate the interface requirements.

Sample Questions for Individual Users/Patients:

- How do you currently identify changes or conditions on your facial skin? (Self-examination / Clinical Consultation / Mobile Apps)
- How long does it usually take to get a professional dermatological evaluation in your area?
- What language would you prefer for an AI-driven health advisory system?
- How comfortable are you with using a laptop webcam or smartphone for real-time facial scanning?
- Have you ever found it difficult to distinguish between a permanent skin feature (e.g., a wrinkle) and a temporary expression?

Appendix B: Detailed Technical Diagrams

As the AI Lead, the following diagrams represent the internal logic and optimization strategies of the framework. These demonstrate the technical depth of the multi-scale diagnostic approach.

- **Real-time Browser-Based Pipeline (MobileNetV2 + WebGL):** A flowchart showing the capture of video frames via the laptop webcam, followed by hardware-accelerated inference using TensorFlow.js to achieve sub-35ms latency.
- **StyleGAN2-ADA Pipeline for Rare Skin Conditions:** A diagram illustrating the feedback loop between the Generator and Discriminator, specifically highlighting how Adaptive Discriminator Augmentation stabilizes training on limited datasets of specific South Asian skin pathologies.
- **INT8 Post-Training Quantization (PTQ) Workflow:** A system diagram showing the conversion of the model from FP32 to 8-bit integers, reducing the footprint to approximately 3.5 MB for efficient edge execution.

Appendix C: Preliminary UI Mockups

The following descriptions outline the visual interface for the specialized diagnostic and management modules.

Patient/User Web Interface (The Client View)

- **Main Scanning Screen:** A clean, browser-based interface featuring a "Facial Alignment Overlay" to guide the user in positioning their face correctly in front of the laptop webcam.
- **Result Screen:** A diagnostic dashboard overlay showing detected conditions confidence scores, and a "Get Expert Advice" button that triggers the personalized health advisory.

Specialist & Security Dashboard (The Professional View)

- **Granular Diagnostic Report:** A detailed view for dermatologists showing high-resolution crops of identified pathologies, historical tracking of skin conditions, and melanin profile analysis.
- **Biometric Integrity Panel:** A strategic view for security analysts displaying "Liveness Detection" metrics and deepfake verification scores to ensure the authenticity of the session.

Management Dashboard (The Healthcare & Security Administrator's View)\

- **Demographic Health Heatmap Screen**: A geospatial visualization of Sri Lanka, with "Hotspots" indicating regions with high prevalence of specific skin pathologies (e.g., Acne clusters or Age Spot trends) identified via the web-based diagnostic tool.

Appendix D: Risk Analysis and Mitigation Matrix

The research identifies several technical and physiological risks specifically related to high-precision facial computer vision and biometric integrity.

Identified Risk	Identified Risk	Mitigation Strategy
Technical	Visual Mimicry: Confusing pathological conditions (Acne) with temporary physiological markers.	Implementing a multi-label reasoning layer that distinguishes between static pathological features and dynamic blood-flow variations.
Data	Data Scarcity: Lack of high-quality labeled images for rare Sri Lankan skin pathologies or specific South Asian phenotypes.	Utilizing a StyleGAN2-ADA pipeline to generate high-fidelity synthetic images for rare classes to achieve dataset balance.
Environmental	Illumination Noise: Variations in indoor lighting (laptop glare) or shadows causing feature distortion on the skin surface.	Utilizing heavy data augmentation during training, including specific lux-level variations and Gaussian noise to simulate diverse home environments.
Operational	Client-Side Latency: High-resolution facial processing causing browser lag or crashing on low-resource hardware.	Applying INT8 Post-Training Quantization (PTQ) to reduce model size to ~3.5MB, ensuring sub-35ms offline inference.